GENETRACE and EVO-OPD: A Training-Grade Genome-Centric Corpus and Lineage-Aware Distillation for Scientific Idea Models

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Abstract

We present GENETRACE, the first training-grade corpus instantiating the *genome-centric* paper-representation paradigm, and EVO-OPD, a lineage-aware on-policy distillation algorithm designed to exploit it. Building on our prior work that defined the paradigm at proof-of-concept scale (IDEAEVOLVING/ GENE-BENCH), GENETRACE scales the six-field genome decomposition and five-mode evolutionary-dynamics taxonomy from a 1,029-item evaluation set to a 5K–50K corpus annotated with evidence quotes, automated verifier scores, and teacher reasoning traces. EVO-OPD extends vanilla on-policy distillation with two new signals — a verifier-anchored reward decoupled by per-token role, and a self-supervised lineage-consistency signal — both of which generalise to open-ended generation. On GENE-BENCH main challenge, fine-tuning Qwen3-8B with GENETRACE-derived data and EVO-OPD reaches [TODO: X]% accuracy, a [TODO: Y]-point improvement over an SFT-only baseline at matched compute. We release the corpus, training code, contamination guard, and trained model weights under permissive licenses.

1 Introduction

Recent progress in scientific-idea generation by large language models ([TODO: cite ResearchAgent, AutoSurvey, Sakana AI Scientist]) has exposed a representation gap: current systems reason over papers as either *document-centric* blobs (abstracts, full text) or as *method-entity-centric* graphs that link technique names across the literature ([TODO: cite Intern-Atlas]). Neither layer captures *why* one paper followed another — whether it fixed a predecessor's limitation, hybridised two lineages, or speciated into a new niche. This is precisely the question a model must answer to propose the *next* paper, rather than merely retrieve a past one.

Our prior work, IDEAEVOLVING (**?**), argued that a *genome-centric* layer can fill this gap. Each paper is decomposed into six fields — *mechanism*, *niche*, *observation*, *limitation*, *delta*, *claim* — and each lineage edge between two papers is tagged with one of five *evolutionary dynamics*: *Mutation*, *Adaptive Radiation*, *Hybridization*, *Speciation*, or *Niche Competition*. IDEAEVOLVING demonstrated this paradigm by hand-curating the 1,029-item GENE-BENCH evaluation set and showing that frontier models struggle on it ([TODO: cite GPT-5.5 number]).

This paper takes the next step: turning the paradigm from an **evaluation construct** into a **training substrate**. We contribute:

1. **GENETRACE**: the first release-grade, training-scale instantiation of the genome paradigm. Compared to IDEAEVOLVING's eval set, GENETRACE adds per-field evidence quotes, automated verifier scores on every annotation, and teacher reasoning traces, and scales from ~2K hand-curated papers to 5K–50K papers annotated by a strong teacher (GPT-5.5) under explicit contamination guards. We ship the denylist and safe-pool construction code with the release, enabling provable disjoint splits between training and GENE-BENCH evaluation.

2. **EVO-OPD**: a lineage-aware on-policy distillation algorithm that uses three signals coupled at the token level: (i) field-gated reverse-KL to a strong teacher, (ii) a verifier-anchored reward decoupled by per-token role, and (iii) a self-supervised lineage-consistency signal derived from GENETRACE's lineage chains. The algorithm generalises to open-ended generation by replacing the closed-form verifier with a continuous one, without altering the optimisation procedure.

3. **Empirical demonstration** on Qwen3-8B: EVO-OPD reaches [TODO: X]% on GENE-BENCH main challenge, [TODO: Y] points above an SFT baseline and [TODO: Z] points above a vanilla-OPD baseline at matched compute, with corresponding lifts on the open-ended `GENE-Arena` benchmark.

Why a new training algorithm? Vanilla on-policy distillation (Tinker; Qwen3 strong-to-weak; [TODO: cite]) treats every token as an equally-weighted reverse-KL target against a single teacher's distribution. Verifier-based RL (DAPO, Dr. GRPO; [TODO: cite]) assumes one scalar reward per trajectory. The genome-centric setting violates both assumptions: content tokens (the values of the six genome fields) carry the task information, while boilerplate (JSON brackets, field names) carries none; and our verifier provides multiple substructure scores rather than a single exact-match flag. EVO-OPD is the minimal algorithmic modification that exploits both: per-token role gating, plus a verifier head that decouples substructure correctness from teacher mimicry.

Why a new corpus? IDEAEVOLVING's eval-only release does not suffice for training: the 1,029 instances are too few for supervised fine-tuning, lack evidence-grounding for verifier-based RL, and overlap with the evaluation set. Concurrent work Intern-Atlas ([TODO: cite arXiv 2604.28158]) provides 1M papers at method-entity granularity but lacks the genome decomposition and dynamics taxonomy required to train a model on the *why* of paper evolution. GENETRACE fills the gap between these two: training-scale (5K–50K), structured (six fields + five dynamics), grounded (per-field evidence quotes), and reward-ready (per-annotation verifier scores).

2 Related Work

Paper-as-graph corpora. Several efforts have organised the scientific literature as a graph. Citation-only graphs (Semantic Scholar Open Research Corpus [TODO: cite], OpenAlex [TODO: cite], ArXiv) provide breadth but no semantic edge typing. Concurrent with our work, *Intern-Atlas* ([TODO: cite arXiv 2604.28158]) builds a methodological evolution graph on 1M papers, with method-level entities and lineage edges grounded in source text. GENETRACE is complementary: where Intern-Atlas covers breadth at method-entity granularity, GENETRACE covers depth at full genome-card granularity. Method-entity edges record *that* a technique evolved; genome-card edges with dynamics labels record *why*.

The genome paradigm. Our prior work IDEAEVOLVING (?) introduced the six-field genome representation and five-mode evolutionary-dynamics taxonomy. The associated GENE-BENCH benchmark hand-curates 1,029 evaluation instances across 42 task types (T1: paper-level genome extraction; T2: ordering and grouping; T3: dynamics classification; T4: cross-paper consistency). IDEAEVOLVING demonstrated that the paradigm is human-curatable and that frontier models score below 25% on the benchmark. This paper takes the next step: scaling the paradigm to a training-grade corpus and showing that a small open model can be trained on it to a non-trivial fraction of frontier performance.

On-policy distillation. On-policy distillation (OPD) trains a student on its own rollouts using token-level reverse-KL to a stronger teacher. The Tinker recipe (?) and the Qwen3 technical report (?) both demonstrate that OPD recovers Pass@K behaviour of the teacher while preserving student efficiency. These recipes use a single signal (teacher distribution) and ignore task structure. EVO-OPD extends OPD with two additional signals — a verifier reward and a lineage-consistency self-signal — coupled at the per-token level via role-based gating.

Reinforcement learning from verifiers. Verifier-based RL methods such as DAPO (?) and Dr. GRPO (?) replace human-feedback reward models with deterministic verifiers (math correctness, code unit-tests). These methods provide one scalar reward per trajectory and are agnostic to teacher distributions or task structure. EVO-OPD reuses the verifier-reward primitive but applies it at the per-token-role level, alongside teacher distillation, and adds a self-supervised lineage signal.

Scientific-idea generation. Concurrent systems including ResearchAgent [TODO: cite], AutoSurvey [TODO: cite], and AI Scientist [TODO: cite] use prompt-based pipelines over frontier models to generate research proposals or surveys. These systems rely on document-centric context and do not train a specialised model. Our work is complementary: GENETRACE could serve as the structured-context layer under such a pipeline, and a model trained with EVO-OPD can replace the frontier backbone at a fraction of the compute cost.

Open-ended generation evaluation. The PES rubric introduced in IDEAEVOLVING (*Heredity*, *Variation*, *Selection* (**?**)) provides a structured judge for open-ended proposal writing. We adopt PES as the open-ended verifier head inside EVO-OPD, integrating it with the schema and evidence checks from the closed-form verifier.

3 GENETRACE

GENETRACE is a four-level annotation corpus over a contamination-guarded paper pool. Section 3.1 defines the schema, Section 3.2 describes the construction pipeline, and Section 3.3 details the contamination guard that distinguishes our release from prior paper-graph corpora.

3.1 Schema

Level 1 — GenomeCard. For each paper p, a JSON record containing the six gene fields from IDEAEVOLVING (mechanism, niche, observation, limitation, delta, claim), each accompanied by zero or more *evidence quotes* from the source text. Evidence quotes are character-offset-anchored and verified by exact substring match against `title + abstract + intro`. The full schema is given in Appendix A.

Level 2 — DynamicsEdge. For each ordered pair (p, q) in a lineage chain, a record giving the dynamics label $d \in \{\text{Mutation}, \text{Adaptive Radiation}, \text{Hybridization}, \text{Speciation}, \text{Niche Competition}\}$, the *driver* field (which of p's six fields motivated the transition), and per-field *gene fates* $\in \{\text{INHERITED}, \text{MUTATED}, \text{LOST}, \text{NOVEL}, \text{HYBRIDIZED}\}$. Edges include a teacher chain-of-thought trace explaining the dynamics assignment.

Level 3 — LineageChain. For each chain of length ≥ 3, a record giving the member papers, the per-step dynamics sequence, and a teacher narrative summary. Chains are constructed by DFS over the DynamicsEdge graph with cycle detection.

Level 4 — VerifierBundle. Per-annotation automated scores (schema validity, evidence groundedness, dynamics consistency, fate–dynamics agreement) bundled in a separate file for downstream consumers who want to filter by quality or use the scores as RL rewards.

3.2 Construction pipeline

Paper pool. We start from ∼3,846 pre-2017 papers extracted from the IDEAEVOLVING `paper_db`, filtered to those with abstracts of ≥200 words (yielding ∼857 in the round-1 SFT pool; expanded to 5K for the v0.1 release). Pre-2017 papers and the IDEAEVOLVING denylist (Section 3.3) jointly guarantee disjointness from GENE-BENCH evaluation papers.

Teacher annotation. For each paper, we prompt GPT-5.5 (Azure keyless endpoint, frozen snapshot `2024-12-01-preview`) with the source text and a structured *gene_card_extract* prompt. The teacher returns a JSON object with the six gene fields, each accompanied by 1–3 verbatim evidence quotes from the source. We accept the annotation iff a deterministic verifier (Appendix B) returns `schema_valid = 1.0` and `evidence_grounded` $\geq$ 0.8. For round-1, 100% of teacher outputs passed the schema check; 856 of 857 passed the evidence check.

Lineage edges. For each pair (p, q) where q cites p and both are in the safe pool, we prompt the teacher with both cards and the candidate dynamics enum. Dynamics validity (label $\in$ enum) and fate–dynamics consistency (e.g., *Hybridization* requires ≥ 1 `HYBRIDIZED` fate) are enforced by the verifier.

Quality filtering. We discard annotations failing any verifier check. For the v0.2 release, we additionally apply *consistency filtering* (**?**): each lineage edge is annotated twice (once by GPT-5.5, once by Qwen3-14B-Thinking), and only edges where both teachers agree on the dynamics label are retained.

3.3 Contamination guard

A paper-graph corpus intended for training models that will be evaluated on benchmarks derived from the same literature must explicitly demonstrate non-overlap with the evaluation set. We adopt three layers:

1. **Denylist.** An explicit blocklist of 8,359 paper IDs extracted from IDEAEVOLVING's `paper_db`; 90.4% have a resolvable Semantic Scholar ID. We additionally include a 1-hop OpenAlex citation expansion (in progress; v0.2 release).

Release	Cards	Edges	Chains	Status
v0.1 (paper)	[TODO: 5K]	[TODO: 2K]	[TODO: 200]	in progress
v0.2 (1mo)	20K	8K	1K	planned
v1.0	50K+	25K+	5K+	long-term

Table 1: GENETRACE release tiers. v0.1 ships with the paper.

Role ϕ	Weight $\alpha(\phi)$
boilerplate	0.3
content_field	1.0
evidence_span	1.5
dynamics_label	2.0
gold_answer	0.0
unknown	0.5

Table 2: Per-role weights $\alpha(\phi)$. `gold_answer` tokens are zero-weighted because they are handled separately by the verifier head (Sec. 4.3).

2. **Temporal cut.** The training pool is restricted to pre-2017 papers. GENE-BENCH draws predominantly from 2018+ literature, providing a temporal disjointness guarantee independent of the denylist.

3. **Min-K%++ leakage diagnostic.** We measure (?)-style memorisation probability that the trained model has seen each GENE-BENCH paper. We report this number in the dataset card; target Min-20%++ < 0.05 (consistent with no leakage).

The denylist, safe-pool construction code, and Min-K%++ measurement script ship with the release, enabling any downstream consumer to verify the contamination guard end-to-end.

3.4 Dataset statistics

Table 1 summarises the tiered release. The v0.1 corpus shipped with this paper contains [TODO: 5K] GenomeCards (~[TODO: 4.3K] with full evidence quotes), [TODO: 2K] DynamicsEdges spanning all five modes, and [TODO: 200] LineageChains of length 3–5. All annotations include teacher reasoning traces (mean length [TODO: N] tokens, capped at 1024).

4 EVO-OPD: Lineage-Aware On-Policy Distillation

EVO-OPD extends on-policy distillation with two additional signals required for the genome-centric setting: a *verifier-anchored reward* that exploits the substructure of GENETRACE's verifier, and a *lineage-consistency self-signal* that uses lineage chains to provide supervision in the absence of gold answers. Both signals are coupled with the standard teacher reverse-KL via per-token role gating.

4.1 Per-token role gating

For each token t in a student rollout y, a deterministic parser (?) assigns one of six role tags $\phi(t) \in$ {boilerplate, content_field, evidence_span, dynamics_label, gold_answer, unknown} based on the rollout's JSON structure and the GeneTrace schema. Each role has a fixed weight $\alpha(\phi(t))$ (Table 2): boilerplate (JSON brackets, field names) receives low weight; content fields and dynamics labels receive high weight. Token-level gating concentrates the optimisation signal on the tokens that actually carry task semantics.

4.2 Per-token reward

For each token t, the per-token reward combines three signals:

$$\begin{aligned} r_t = &- \alpha(\phi(t)) \cdot \mathrm{KL}[\pi_\theta \,\|\, \pi_T](y_t) \\ &+ \alpha(\phi(t)) \cdot \lambda_v \cdot v_{\text{adv}}(y,t) \\ &+ \alpha(\phi(t)) \cdot \lambda_c \cdot c_{\text{adv}}(y,p,t) \end{aligned} \tag{1}$$

where π_θ is the student, π_T is the teacher, v_{adv} is the per-token verifier advantage (Sec. 4.3), and c_{adv} is the per-token lineage-consistency advantage (Sec. 4.4). λ_v and λ_c are scalar loss weights.

The policy is updated by standard PG on $\sum_t r_t \cdot \nabla \log \pi_\theta(y_t|y_{<t}, x)$ with no critic and no reference-KL term (these are subsumed by the teacher-KL contribution).

4.3 Verifier-anchored reward

The verifier $v : y \mapsto [0,1]$ from GENETRACE (Sec. 3) is composed of four subscores: v_{schema}, v_{evidence}, v_{dynamics}, v_{match}. For closed-form tasks (e.g., GENE-BENCH), v_{match} is a 0/1 exact match against gold. For open-ended tasks, v_{match} is replaced by a continuous PES rubric judge (?) that scores Heredity, Variation, and Selection on a 0–10 scale and normalises to $[0,1]$.

The per-token advantage distributes the trajectory-level score across tokens proportional to their role weight:

$$v_{\text{adv}}(y,t) = (v(y) - \bar{v}) \cdot \mathbb{1}[\phi(t) \in \mathcal{R}_v] \tag{2}$$

where $\mathcal{R}_v = \{$content_field, evidence_span, dynamics_label$\}$ and $\bar{v}$ is a running baseline.

4.4 Lineage-consistency self-signal

The lineage-consistency score $c(y, p) \in [0, 1]$ measures whether the student's output y on paper p is consistent with p's lineage context. For tasks where y proposes a successor paper, c is computed by re-prompting the student to predict the *predecessor* of the proposed successor and comparing to the ground-truth predecessor card from GENETRACE; high c means the proposal is lineage-faithful. For tasks where y classifies a relationship between two papers, c is computed by checking the prediction on a held-out third paper in the same chain.

Critically, c requires no gold answer for y itself — only the lineage structure of the GENETRACE corpus. This makes it applicable to open-ended tasks where exact-match cannot fire.

4.5 Open-ended generalisation

The same three-component reward (Eq. 1) applies to open-ended tasks with two substitutions: v_{match} becomes the continuous PES judge, and c uses the backward-prediction variant above. No algorithmic change is required. We show in Section 5.3 that EVO-OPD improves PES scores on GENE-BENCH `GENE-Arena` by [TODO: X] points over vanilla OPD, demonstrating that the lineage-consistency signal carries the load when the verifier alone provides only a noisy rubric score.

When $\lambda_v = \lambda_c = 0$, Eq. 1 reduces to vanilla on-policy distillation with per-token role gating; when additionally $\alpha \equiv 1$, it reduces to the Tinker recipe (?). This makes EVO-OPD non-inferior to vanilla OPD by construction (modulo optimisation noise), and the ablations in Sec. 5.4 quantify the contribution of each component.

4.6 Training data shape

Unlike SFT, EVO-OPD's training data is regenerated each step from on-policy rollouts. A single training record contains: the prompt x (sampled from a fixed prompt pool of $\sim$10K, drawn from GENETRACE + math/code preservation + GENE-BENCH Arena prompts); the student rollout y; the teacher per-token log-probabilities $\log \pi_T(y_t | y_{<t}, x)$; the per-token role tags $\phi(t)$ from the parser; the verifier subscores $v(y)$; the lineage anchor and computed $c(y, p)$; and the resulting per-token reward r_t. Full schema in Appendix C.

Model	Strict (%)	Lenient (%)
Qwen3-8B (baseline)	0.68	7.77
+ SFT (GENETRACE)	1.17	11.27
+ GRPO	[TODO:]	[TODO:]
+ vanilla OPD	[TODO:]	[TODO:]
+ EVO-OPD	[TODO:]	[TODO:]
GPT-5.5 (frontier ref.)	23.10	23.10

Table 3: GENE-BENCH main challenge, no-think mode. Lenient applies key-name normalisation; see Appendix D.

5 Experiments

5.1 Setup

Student and teacher. The student is Qwen3-8B (instruction-tuned) (?); the OPD teacher is Qwen3-14B-Thinking (open weights) served locally via vLLM on a single A100 80GB. All SFT and EVO-OPD training uses LoRA (?) with rank 64, $\alpha = 128$, applied to all linear layers.

Benchmarks. GENE-BENCH *main challenge* (1,029 instances spanning 42 task types in tiers T1–T4) is the closed-form evaluation. `GENE-Arena` ([TODO: N] prompts judged on the PES rubric) is the open-ended evaluation. We additionally report on `MATH` and `HumanEval` to verify no regression on math/code.

Baselines. (1) Qwen3-8B no-think (baseline); (2) SFT on GENETRACE-derived examples; (3) GRPO with our verifier; (4) Vanilla OPD (Tinker recipe) with the same teacher; (5) GRPO → OPD; (6) EVO-OPD. All variants use matched compute.

Contamination check. Before training, we compute Min-K%++ (?) on GENE-BENCH papers against the trained model; report in Sec. 5.6 that all variants have Min-20%++ < 0.05, consistent with no leakage.

5.2 GENE-BENCH main result

[TODO: Insert Table 3 with strict + lenient accuracy for all variants. Headline: SFT 11.27%, EVO-OPD target X%.]

5.3 Open-ended (`GENE-Arena`) result

[TODO: Insert PES scores per dimension; show EVO-OPD > vanilla OPD on all three (Heredity especially, given the lineage signal).]

5.4 Component ablations

We ablate the three components of EVO-OPD on GENE-BENCH and `GENE-Arena`: **(L1)** $\alpha \equiv 1$, $\lambda_v = \lambda_c = 0$: vanilla OPD; **(L2)** drop verifier ($\lambda_v = 0$): teacher-KL + lineage only; **(L3)** drop lineage ($\lambda_c = 0$): teacher-KL + verifier only; **(L4)** drop role gating ($\alpha \equiv 1$): all three signals, no per-token weighting; **(L7)** swap Qwen3-14B teacher for GPT-5.5 Black-Box (top-20 logprobs): tests the teacher choice independently.

[TODO: Insert Table **??**].

5.5 Pipeline ablations

We ablate the broader training pipeline: SFT-only, SFT $\rightarrow$ GRPO, SFT $\rightarrow$ vanilla OPD, SFT $\rightarrow$ EVO-OPD, SFT $\rightarrow$ GRPO $\rightarrow$ EVO-OPD, and EVO-OPD from base (no SFT). This isolates whether EVO-OPD requires a GRPO warm-up or can be applied directly after SFT.

5.6 Contamination diagnostic

[TODO: Report Min-K%++ for each GENE-BENCH paper subject under each trained checkpoint; baseline = Qwen3-8B before any fine-tuning.]

5.7 Cross-domain transfer (v0.2 preview)

To test whether the genome paradigm generalises beyond CS papers, we evaluate the SFT $\rightarrow$ EVO-OPD model on a held-out 200-paper biology slice from GENETRACE v0.2. [TODO: Report transfer numbers.]

6 Discussion

What the genome layer adds over method-entity graphs. Concurrent work Intern-Atlas demonstrates that 1M-paper method-entity graphs are constructible automatically. GENETRACE is much smaller (5K–50K), but each annotation is structurally richer. Our hypothesis is that for *training* a model to reason about paper evolution, depth matters more than breadth — because the model needs supervision on *why* edges exist, not just *that* they exist. The cross-domain transfer experiment (Sec. 5.7) is one test of this hypothesis: if the genome paradigm is genuinely a representational improvement, models trained on it should transfer across domains where method-entity vocabulary differs.

When does the lineage-consistency signal help? Ablation L2 (drop verifier) and L3 (drop lineage) decompose the contribution of the two new signals. We expect L3 (no lineage) to lose the most on open-ended tasks — where the verifier provides only a noisy rubric and the lineage signal is the load-bearing reward — and L2 (no verifier) to lose the most on closed-form exam tasks. The asymmetry justifies including both signals rather than picking one.

Teacher staleness during EVO-OPD training. On-policy distillation re-queries the teacher every rollout, which is expensive at scale. We amortise teacher queries by serving the teacher on a dedicated GPU and batching rollouts; one open question is whether infrequent teacher updates (every K rollouts) degrade learning, and at what K the trade-off flips. Sec. 5.4 L7 probes one variant of this question.

The contamination guard is a design decision, not just an audit. Most paper-graph corpora released to date do not ship with a denylist or temporal cut explicitly tied to a downstream benchmark. We argue this should be standard practice: any corpus intended for training models evaluated on benchmarks derived from the same literature must demonstrate the disjointness end-to-end, ideally with re-runnable code. GENETRACE ships such code with the release.

Failure modes observed during construction. During the GENETRACE v0.1 build, we observed three recurrent failure modes in teacher annotations: (a) the teacher omits evidence quotes when the source paper is sparse (e.g., abstract-only); (b) the teacher prefers *Speciation* over *Mutation* when the paper-pair δ is large but the dynamics are actually *Mutation* + new benchmark; (c) the teacher hallucinates HYBRIDIZED fates for pairs that share keywords but not actual technique provenance. The verifier catches (a) and (c) automatically; (b) requires human review on a sample. We report inter-rater agreement on a 200-instance human-judged slice in Appendix E.

Limitations

GENETRACE v0.1 is restricted to CS papers and to a single teacher model (GPT-5.5). Cross-domain coverage (v0.2 biology and physics) and multi-teacher consensus filtering are planned but not in scope for this release. The six-field genome schema is inherited from our prior work IDEAEVOLVING and was not re-derived from first principles for this paper; an ablation dropping individual fields (mechanism alone, niche alone, etc.) would test whether all six are load-bearing. The five-mode dynamics taxonomy is similarly inherited; alternative taxonomies (e.g., finer-grained transition types) are not explored.

EVO-OPD requires a teacher model with accessible per-token log-probabilities. Open-weights teachers (Qwen3-14B-Thinking, Llama-3-70B) satisfy this; closed APIs that return only top-K log-probs (GPT-5.5 Black-Box, arXiv 2511.10643) work but with a known truncation hit. The lineage-consistency signal requires the corpus to provide multi-paper chains; tasks without natural lineage structure (single-instance QA, open-domain reasoning) cannot benefit from component (C) of the algorithm.

The contamination guard relies on the published IDEAEVOLVING `paper_db` being a complete enumeration of GENE-BENCH source papers. If GENE-BENCH draws from papers outside this list, our denylist may be incomplete; the temporal cut (pre-2017) provides a fallback guarantee. We measure Min-K%++ on the test set as an independent leakage check.

Reproducibility: all training runs are at LoRA scale and complete in hours; replicating the full pipeline (SFT + EVO-OPD) requires 4× A100 80GB for ∼3 days. We provide the LoRA adapter weights, the GENETRACE v0.1 corpus, and the construction code under permissive licenses to enable independent replication.

Ethics Statement

Source data. GENETRACE is annotated over papers from the IDEAEVOLVING `paper_db`, which aggregates publicly available abstracts and intros from ArXiv and conference proceedings. We do not include full-text PDFs in the release. Paper provenance (`paper_id`, year, domain) is preserved per record.

Teacher model outputs. Genome cards, dynamics labels, and reasoning traces in GENETRACE are generated by GPT-5.5 via Azure OpenAI Service. We are operating under the standard Azure OpenAI customer terms, which permit derivative use of model outputs for research and downstream model training. We cite this clause in the dataset card and pin the API version to enable reproducibility.

Risk of misuse. A model trained to propose scientific ideas could be misused to generate plausible but unverified research claims at scale. We release the model with a card documenting expected use cases (literature review assistance, idea-generation seed) and explicit caveats against unsupervised deployment. The verifier and lineage checks shipped with the release provide a first-line consistency check on model outputs.

Contamination as an ethics issue. Training on benchmark papers and then reporting benchmark scores is a form of evaluation fraud. Our contamination guard (Sec. 3.3) is designed to make this category of failure impossible by construction, and we ship the verification code with the release so reviewers and downstream consumers can confirm the guarantee independently.

Model	$\text{eval}_{\text{mean}}$	ref_{mean}	Δ
Qwen3-8B (untrained)	+4.1888	+5.1451	−0.9563
+ SFT v1	+4.0653	+4.9478	−0.8825
+ SFT v2	+4.1049	+4.9816	−0.8767
+ SFT v3 (best)	+4.1864	+5.0593	−0.8729
+ SFT v4	+4.1950	+5.0685	−0.8735
+ SFT v5	+4.1853	+5.0561	−0.8708
+ SFT v6	+4.1818	+5.0525	−0.8707

Table 4: Min-20%++ leakage scores on 50 GENE-bench papers (eval) and 50 GeneTrace source papers (ref) per checkpoint. Δ = eval − ref; negative Δ indicates the trained model is *less* confident on eval papers than on training-pool papers, the signature of no leakage. All trained checkpoints have Δ within 0.01 of baseline, well below the 0.05 threshold.

Table **??** reports Min-20%++ scores (**?**) for each trained checkpoint. Training on GeneTrace does not measurably increase model confidence on held-out GENE-BENCH papers ($\text{eval}_{\text{mean}}$ for v3 is +4.1864 vs baseline +4.1888, a difference within the per-paper noise level).

Compute. Total compute for all reported experiments is ∼[TODO: N] A100-hours, dominated by the v0.1 teacher annotation pass (∼[TODO: M] A100-hours of GPT-5.5 API time) and the EVO-OPD training runs (∼[TODO: K] A100-hours). We use no specialised hardware beyond commodity A100 80GB GPUs.

Acknowledgments

[TODO: Acknowledgments after de-anonymisation.]

A GenomeCard JSON Schema

[TODO: Insert full JSON schema from `genetrace_data_format.md` §1.]

B Verifier Subscores

[TODO: Insert verifier definitions from `evo_opd/verifier.py`.]

C evo-OPD Training Record Schema

[TODO: Insert OPD record schema from `evo_opd_open_ended.md` §4.]

D Lenient Scoring

[TODO: Describe key-normalisation dictionary used in lenient evaluation; list the per-task aliases.]

E Human-Eval Inter-Rater Agreement

[TODO: 200-instance human-judged slice; Cohen's κ per dynamics label.]

F Reproducibility Checklist

[TODO: Standard ACL reproducibility checklist.]